Team 7

Project Title: Local Governments Survey (Harmonizing ARISE City Survey + CivicPulse Kansas Sample)

Course Number:CS896

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ARISE DATASET AND CODEBOOK EXPLAINATION –

The **ARISE dataset** is a survey-based dataset collected from Kansas city officials to understand how local governments deal with infrastructure, hazards, and financial challenges. Each row in the dataset represents one city official's response, and each column represents a survey question, such as concerns about floods, electricity systems, workforce shortages, or overall fiscal condition.

The data is mostly stored as numbers (for example, 1–5 scales or ranked hazards), and those numbers represent specific meanings explained in the codebook. In simple terms, the ARISE dataset shows how Kansas cities perceive risks, manage infrastructure, and prepare for disasters.

It contains 309 responses (Rows) & 91 Variables (Columns).

The **ARISE codebook** is the guide that helps us understand the dataset. It explains what each column name means and what each number represents. For example, if a column shows a value of 5, the codebook tells us whether that means Extremely concerned, Excellent, or something else. Without the codebook, the dataset would just look like random numbers and short variable names like Q3_12 or Q119.

CODEBOOK

COUNTIES		The name of the respondent's county
Urban		Located in Metropolitan counties (1) Located in Micropolitan or Rural counties (0)
Q3_1st.Dis	Q3 Which of these hazards have had the greatest impact on your city over the past five years? (Identify up to three by dragging the relevant hazards into the box.) • Agricultural infestation/disease (10) • Cyberattack (9) • Dam or levee failure (19) • Droughts (1) • Earthquake (12) • Extreme heat (2) • Floods (3) • Ice/winter storms or hails (4) • Hazardous waste spill/release (7) • Industrial or commercial fire (8) • Soil erosion and dust (22) • Severe storms/high winds (21) • Terrorism/Agro-terrorism (17) • Tornados (5) • Utility/Infrastructure failure (11) • Wildfires (6) • Other: (23)	• Code of the 1st ranked disaster • Did not respond (NA)
Q3_2nd.Dis		• Code of the 2nd ranked disaster • Did not respond (NA)
Q3_3rd.Dis		• Code of the 3rd ranked disaster • Did not respond (NA)
Q3_10_Agri		• Agricultural infestation/disease is ranked (1), (2), or (3) • If not ranked in the top 3 (NA)
Q3_9_Cyber		Same as Q3_10_Agri
Q3_19_Dam		
Q3_1_Droughts		
Q3_12_Earthquake		
Q3_2_heat		
Q3_3_Floods		
Q3_4_Ice		
Q3_7_waste.spill		
Q3_8_Industrial.fire		
Q3_22_Soil.erosion		
Q3_21_Severe.storms		
Q3_17_Terrorism		
Q3_5_Tornados		
Q3_11_Infrastructure.failure		
Q3_6_Wildfires		
Q3_23_Others		
Q3_23_TEXT		• The text coming with Q3_23_Others • Did not respond (NA)

DATASET

G1 | 3rd Dis

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	
1		ResponseI	RecipientL	ExternalRe	1st Dis	2nd Dis	3rd Dis	Q3_10_Agr	Q3_9_Cybe	Q3_19_Dai	Q3_1_Drou	Q3_12_Ear	Q3_2_heat	Q3_3_Floo	Q3_4_Ice	Q3_7_wast	Q3_8_Indu	Q3_22_Soi	Q3_21_Sev	Q3_17_Ter	Q3_5_Torn	Q3_11_Infr	Q3_6_Wild	Q3
2	1	R_2sQNn7	chief admi	2079950	3	21	1	NA	NA	NA	3	NA	NA	1	NA	NA	NA	NA	2	NA	NA	NA	NA	NA
3	2	R_2Qyp1Yf	chief admi	2061250	11	3	8	NA	NA	NA	NA	NA	NA	2	NA	NA	3	NA	NA	NA	NA	1	NA	NA
4	3	R_2aQAihY	chief admi	2005600	11	21	1	NA	NA	NA	3	NA	NA	NA	NA	NA	NA	NA	2	NA	NA	1	NA	NA
5	4	R_6sTVkGI	chief admi	2034300	21	1	2	NA	NA	NA	2	NA	3	NA	NA	NA	NA	NA	1	NA	NA	NA	NA	NA
6	5	R_3sddVtn	chief admi	2053225	3	1	23	NA	NA	NA	2	NA	NA	1	NA	NA	NA	NA	NA	NA	NA	NA	NA	
7	6	R_6rJcm8c	chief admi	2032575	1	2	21	NA	NA	NA	1	NA	2	NA	NA	NA	NA	NA	3	NA	NA	NA	NA	NA
8	7	R_3PUjNBI	chief admi	2007650	21	4	5	NA	NA	NA	NA	NA	NA	NA	2	NA	NA	NA	1	NA	3	NA	NA	NA
9	8	R_2OGOtC	chief admi	2035750	11	21	1	NA	NA	NA	3	NA	NA	NA	NA	NA	NA	NA	2	NA	NA	1	NA	NA
10	9	R_1LLOnAt	chief admi	2020075	2	6	21	NA	NA	NA	NA	NA	1	NA	NA	NA	NA	NA	3	NA	NA	NA	2	NA
11	10	R_3ekoelh	Planning	2053725	21	2	5	NA	NA	NA	NA	NA	2	NA	NA	NA	NA	NA	1	NA	3	NA	NA	NA
12	11	R_1EgJ0DI	public wor	2077500	21	11	2	NA	NA	NA	NA	NA	3	NA	NA	NA	NA	NA	1	NA	NA	2	NA	NA
13	12	R_xgbCMrr	Planning	2036000	1	22	21	NA	NA	NA	1	NA	NA	NA	NA	NA	NA	2	3	NA	NA	NA	NA	NA
14	13	R_2cinsLh	Planning	2052275	21	4	3	NA	NA	NA	NA	NA	NA	3	2	NA	NA	NA	1	NA	NA	NA	NA	NA
15	14	R_1q9AiNy	Planning	2002300	21	3	4	NA	NA	NA	NA	NA	NA	2	3	NA	NA	NA	1	NA	NA	NA	NA	NA
16	15	R_2EGj2Df	Planning	2043950	21	1	2	NA	NA	NA	2	NA	3	NA	NA	NA	NA	NA	1	NA	NA	NA	NA	NA
17	16	R_2OHH6y	Planning	2039825	4	21	3	NA	NA	NA	NA	NA	NA	3	1	NA	NA	NA	2	NA	NA	NA	NA	NA
18	17	R_yWmAn	public wor	2053450	21	11	1	NA	NA	NA	3	NA	NA	NA	NA	NA	NA	NA	1	NA	NA	2	NA	NA
19	18	R_1Clkurty	public wor	2002675	11	21	1	NA	NA	NA	3	NA	NA	NA	NA	NA	NA	NA	2	NA	NA	1	NA	NA
20	19	R_eS7LrUy	chief admi	2015200	21	4	2	NA	NA	NA	NA	NA	3	NA	2	NA	NA	NA	1	NA	NA	NA	NA	NA
21	20	R_1CJUCA	Planning	2024000	21	4	1	NA	NA	NA	3	NA	NA	NA	2	NA	NA	NA	1	NA	NA	NA	NA	NA
22	21	R_2aslKh4	Planning	2039350	21	5	9	NA	3	NA	NA	NA	NA	NA	NA	NA	NA	NA	1	NA	2	NA	NA	NA
23	22	R_324Q6P	public wor	2022700	3	21	4	NA	NA	NA	NA	NA	NA	1	3	NA	NA	NA	2	NA	NA	NA	NA	NA
24	23	R_1eFgfIN	chief admi	2017800	9	21	3	NA	1	NA	NA	NA	NA	3	NA	NA	NA	NA	2	NA	NA	NA	NA	NA
25	24	R_3PFfWb	public wor	2018500	1	21	2	NA	NA	NA	1	NA	3	NA	NA	NA	NA	NA	2	NA	NA	NA	NA	NA
26	25	R_24vo6pV	chief admi	2002375	21	2	1	NA	NA	NA	3	NA	2	NA	NA	NA	NA	NA	1	NA	NA	NA	NA	NA
27	26	R_1gGzUrz	chief admi	2076475	1	3	11	NA	NA	NA	1	NA	NA	2	NA	NA	NA	NA	NA	NA	NA	3	NA	NA

EXPLANATION WITH EXAMPLE

- The ARISE dataset is a survey of Kansas city officials about hazards, infrastructure, and local government challenges.
- Each row in the dataset represents one respondent, and each column represents a survey question.
- Question 3 asks respondents to rank the top three hazards that have had the greatest impact on their city over the past five years.
- Columns like **1st Dis, 2nd Dis, and 3rd Dis** store hazard identification codes, not rating scales.
- In the codebook, each number represents a specific hazard, such as 3 for Floods, 1 for Droughts, and 21 for Severe storms.
- **1st Dis = 3**, it means the respondent ranked Floods as the most impactful hazard.
- The dataset also stores this ranking information in another format using columns like **Q3_12_Earthquake** or **Q3_3_Floods**.
- In these columns, values 1, 2, or 3 represent whether that hazard was ranked first, second, or third.
- A value of NA means the hazard was not selected in the top three.
- The codebook is important because it explains what each number means and how each question is structured.Without the codebook, the dataset would only show numbers, but with it, we can correctly interpret and analyze the responses.

For this question Q3 the scale is 1-3, every question have a different scale or choices.

This question contain two types of number data - Hazard codes (like 12 = Earthquake)
- Ranking position (1st, 2nd, 3rd)

How much impactful the hazard was

1 → Ranked 1st
2 → Ranked 2nd
3 → Ranked 3rd
NA → Not selected in top 3
So:
If Q3_12_Earthquake = 1
→ Earthquake was ranked as the most impactful hazard.
If Q3_12_Earthquake = 2
→ Earthquake was ranked second.
If Q3_12_Earthquake = NA
→ Earthquake was not chosen in the top three.

CIVICPULSE CODEBOOK EXPLANATION

- The CivicPulse codebook explains every variable included in the survey.
- It is organized into columns such as Field, Field Type, and Description.
- The Field column shows the variable name.
- The Field Type column indicates whether it is Metadata or Response data.
- Metadata fields provide background information about the respondent, such as their role, department, state, or county characteristics.
- Response data fields contain the actual answers given by respondents to survey questions.
- The codebook clearly explains the wording of each survey question.
- It also defines what each numeric response means (for example, 1 = Low condition and 5 = High condition).
- The codebook may also explain how missing values are recorded.
- Overall, the CivicPulse codebook acts as an instruction manual that helps us correctly interpret and analyze the survey data.

MAIN CODEBOOK

	A	B	C
1	**Field**	**Field Type**	**Description**
2	Respondentid	Metadata	A random number generated to uniquely identify each survey respondent.
3	StartDate	Metadata	The date and time when the respondent started the survey.
4	Roletype	Metadata	The type of role in which the survey respondent serves.
5	Govtype	Metadata	The type of government in which the survey respondent serves.
6	Department	Metadata	The department of the survey respondent. This field is 'NA' for County/Municipal Policymakers.
7	Stateabb	Metadata	The state in which the survey respondent serves, provided as the state's two-letter abbreviation.
8	PopulationcountyACS2023_3	Metadata	The total number of residents living in the county in which the respondent serves. This data comes from the 2019-2023 US Census American Community Survey and is sorted into the following bins: <40k, 40k to 160k, >160k.
9	CollegepropcountyACS2023_3	Metadata	The proportion of residents with a 4-yr college degree in the county associated with the respondent. This data is from the 2019-2023 US Census American Community Survey. The data is sorted into the following bins: <0.13, 0.13 to 0.21, >0.21.
10	Voteproprepcounty2024_3	Metadata	The proportion of the votes in the county that went to Donald Trump in the 2024 Presidential election. The data is sorted into the following bins: <55%, 55% to 70%, >70%.
11	Weight	Metadata	Probability weights created with a post-stratification raking procedure using the county-level Census and presidential vote share variables. This procedure follows the methodology outlined in DeBell and Krosnick (2009) for the American National Elections Study (ANES). Weights are calculated separately for each role type.
12	Workforce_SC	Response data	Over the past five years, how challenging has it been for your local government to maintain a sufficient workforce to provide its desired level of services?
13	Challenges_MS	Response data	What are the main reasons it has been challenging to maintain a sufficient workforce? Select all that apply. - Selected Choice
14	Challenges_MS_other_TEXT	Response data	What are the main reasons it has been challenging to maintain a sufficient workforce? Select all that apply. - Other (Please specify): - Text
15	Condition_SC	Response data	If we define "fiscal condition" as a government's capacity to meet short- and long-term financial and service obligations, how would you rate your local government's overall fiscal condition over the past five years?
16	Infra_GR_spending	Response data	Please indicate your level of agreement with the following statements about infrastructure financing in your local government. - Spending has not kept pace with community needs.

There are three columns:

1.Field

This is the variable name (how it appears in the dataset).
Example:
Respondentid
StartDate
Roletype
Workforce_SC
Condition_SC
These are the column names in the dataset.

2.Field Type

This tells you what kind of variable it is.
You can see two main types:
-Metadata
This is background information about the respondent.
It is NOT an opinion answer.
-Response data
This is the actual answer given by the respondent.
Examples:

3. Description

Description is overall explanation of what questions asked and what wording used.

Examples
Workforce_SC
Field Type: Response data

Description: How challenging has it been to maintain a sufficient workforce?”
This means: This variable stores the respondent’s answer to that question.
Likely scale: 1–5 (for example: Not challenging → Extremely challenging)
The codebook is telling you exactly what the question wording was.

Condition_SC
Field Type: Response data
This asks: How would you rate your local government's overall fiscal condition?”
Again: This is a scaled question (likely 1–5).
The codebook helps you understand what the scale means.

HIGH – LOW CODEBOOK

A1 | HLVersion_QR

	A	B	C	D
1	HLVersion_QR	Priority Programming Field	Priority Programming Field Value	
2	1	HL1Priority1_QD	“Electricity”	
3	1	HL1Priority2_QD	“Medications and medical equipment”	
4	1	HL1Priority3_QD	“Safe and secure shelter”	
5	1	HL1Priority4_QD	“Transportation (e.g., open roadways, public transportation)”	
6	1	HL2Priority1_QD	“Supplies or workers to assist with clean up”	
7	1	HL2Priority2_QD	“Broadband internet and cell phone service”	
8	1	HL2Priority3_QD	“Childcare”	
9	1	HL2Priority4_QD	“Financial resources (e.g., cash, credit card services, savings)”	
10	1	HL3Priority1_QD	“Critical medical services (e.g., emergency room)”	
11	1	HL3Priority2_QD	“Safe drinking water”	
12	1	HL3Priority3_QD	“Law enforcement/public safety services”	
13	1	HL3Priority4_QD	“Food supply”	
14	2	HL1Priority1_QD	“Childcare”	
15	2	HL1Priority2_QD	“Broadband internet and cell phone service”	
16	2	HL1Priority3_QD	“Safe drinking water”	
17	2	HL1Priority4_QD	“Transportation (e.g., open roadways, public transportation)”	
18	2	HL2Priority1_QD	“Electricity”	
19	2	HL2Priority2_QD	“Law enforcement/public safety services”	
20	2	HL2Priority3_QD	“Medications and medical equipment”	
21	2	HL2Priority4_QD	“Financial resources (e.g., cash, credit card services, savings)”	
22	2	HL3Priority1_QD	“Supplies or workers to assist with clean up”	
23	2	HL3Priority2_QD	“Critical medical services (e.g., emergency room)”	
24	2	HL3Priority3_QD	“Safe and secure shelter”	
25	2	HL3Priority4_QD	“Food supply”	
26	3	HL1Priority1_QD	“Supplies or workers to assist with clean up”	
27	3	HL1Priority2_QD	“Critical medical services (e.g., emergency room)”	

Here you can see:

- HLVersion_QR
- Priority Programming Field
- Priority Programming Field Value

This table is explaining how ranking questions are structured.
Example:
HL1Priority1_QD → “Electricity”
HL1Priority2_QD → “Medications and medical equipment”
HL1Priority3_QD → “Safe and secure shelter”
What this means:

This is a priority ranking question.
Respondents ranked different community needs like:

- Electricity
- Safe drinking water
- Transportation
- Law enforcement
- Food supply
- Broadband internet

HL1Priority1_QD means:
→ First priority in section 1
HL1Priority2_QD means:
→ Second priority

So this table shows:
How ranked choices are labeled in the dataset.

WEEK 5 GOAL –

Identify the overlapping variables and make a final Crosswalk Table.